

The container derivative is a tangent structure: a fibrational positioning

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3 October 2026 (revised 7 October 2026: scope stated as finite-support; the directed-container restriction relabelled a proposition-sketch, per a referee report)

Abstract

The derivative of a container is usually sold as a pun: $\partial(y^n) = n \cdot y^{n-1}$ is the type of one-hole contexts, and the resemblance to calculus is left as a wink. We remove the wink. We show that the category $\mathbf{SPoly}$ of (finite-support, multivariable) polynomial functors is a Cartesian differential category whose differential combinator is exactly the Abbott–Altenkirch–Ghani–McBride one-hole-context derivative ∂ ; consequently ∂ is the vertical part of a genuine Cartesian tangent structure, realised on the nose by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$. The proof is a reflection along the faithful counting functor into the classical Cartesian differential category of $\mathbb{N}$ -polynomials, so every differential axiom is inherited rather than re-checked. We isolate the one place this picture is routinely misread: the non-trivial content lives at the level of *maps*, not objects — a map $1 \rightarrow 1$ is differential-linear iff it is a linear container $S \cdot y$, so y^2 is not linear and the tangent functor is *not* the trivial $T(p) = p \times p$. We then position the result against the fibrational theory of first-order differential structures, flag the orientation hazard that the container bifibration lives on the *reverse*, lens side of that theory, and record a first restriction result: T is a strict $\triangleleft$ -monoidal base-change functor from directed containers over k to directed containers over the dual numbers, so the tangent bundle of a small category lives over $k[\varepsilon]/\varepsilon^2$.

1 Introduction

The pun, and what is under it. In the theory of containers one learns early that data types have derivatives. If a container’s extension is the polynomial functor y^n — “ n -tuples” — then its one-hole contexts form the functor $n \cdot y^{n-1}$: choose which of the n slots is the hole, and the remaining $n - 1$ slots still carry data. Abbott, Altenkirch, Ghani and McBride made this precise for all containers [1]: the one-hole-context construction ∂ acts on a container $S \triangleleft P$ by, for each shape, replacing one position by a hole, and it satisfies the sum, product and chain rules of elementary calculus. The notation ∂ is suggestive, and the suggestion is usually left as exactly that — a mnemonic that the Leibniz rule $\partial(F \times G) = \partial F \times G + F \times \partial G$ happens to hold for data types.

This note argues that the mnemonic is a theorem. The resemblance to calculus is not a coincidence of low-degree arithmetic; it is the shadow of a *Cartesian differential category* structure [3] in which ∂ is literally the differential combinator. Once that is established, a second structure comes for free: every Cartesian differential category is a Cartesian tangent category [5], so ∂ is the vertical (fibre-direction) part of a genuine tangent bundle functor T , and T is computed by the oldest trick in differential algebra — substitute a dual number and read off the ε -coefficient.

The gap. Two bodies of work sit on either side of this statement without quite touching. On one side, ∂ itself: defined and shown to obey the calculus rules [1], with a chain rule under composition of containers proved explicitly [7], but never (to our knowledge) organised into the axioms of a differential category. On the other side, the fibrational theory of first-order differential structures of Capucci–Cruttwell–Ghani–Zanasi [4], which packages Cartesian differential categories, tangent categories and their reverse variants uniformly as sections of a fibration — but whose running examples are manifolds and lenses, not containers, even though it cites the containers literature. The missing sentence is the one that says: here is a differential category whose combinator is the container derivative, and here is where it sits in the fibrational picture.

The result. Let $\mathbf{SPoly}$ be the category of finite-support multivariable polynomial functors up to isomorphism, with substitution as composition (Definition 2). Our main theorem is the following.

Theorem 1 (Main theorem). *$\mathbf{SPoly}$ is a Cartesian differential category whose differential combinator is the AAGM one-hole derivative ∂ . Consequently $(\mathbf{SPoly}, \partial)$ is a Cartesian tangent category with tangent functor $T(n) = 2n$, $T(f) = \langle f \circ \pi_1, \partial[f] \rangle$, realised by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$.*

The reader meets the precise statement again, with all terms defined, as Theorem 4 on page 5.

The method. We do not verify the seven differential-category axioms by hand on containers. Instead we exhibit a *counting functor* $N: \mathbf{SPoly} \rightarrow \mathbf{Poly}_{\mathbb{N}}$ into the classical Cartesian differential category of polynomials over the rig $\mathbb{N}$, sending a container to the polynomial that counts its elements (Definition 3). We show N preserves all the differential-category structure — crucially $N(\partial_i F) = \partial N(F)/\partial m_i$ — and that N is *faithful* on isomorphism classes. A faithful functor reflects equalities of parallel maps, so each axiom, true in $\mathbf{Poly}_{\mathbb{N}}$ because polynomials over a rig form a Cartesian differential category [3], descends to $\mathbf{SPoly}$. The whole proof is three short lemmas (§3). Reflection is the right tool precisely because the axioms are equations between parallel morphisms, and that is exactly what faithful functors see.

The correction that makes it non-trivial. There is a tempting but vacuous reading of “differential structure on containers” at the level of objects. In a Cartesian differential category one can ask which objects B are *linear*, meaning $TB \cong B \times B$; but $\mathbf{SPoly}$ is a *full* Cartesian differential category, so every object is linear and the question says nothing. The content is one level down, at *maps*. We prove (Proposition 9) that a map $1 \rightarrow 1$ is differential-linear — equivalently additive, equivalently $\partial[f] = f \circ \pi_2$ — if and only if it is a linear container $S \cdot y$. So y is linear but $y^2, y^3, \dots$ are not. This is the certificate that the tangent functor is genuinely differential and *not* the trivial $T(p) = p \times p$ (under which every map would be linear). We state the discriminator as a table (Table 1) because it is the single most useful thing to carry away.

Where it sits, and one hazard. The fibrational theory [4] presents a first-order tangent structure as a section of the *additive codomain fibration*, with reindexing by *pullback*. We record in §5 that our dual-number T realises exactly this forward picture — and we flag, because the programme it belongs to has been bitten by this three times, that the container bifibration $\mathbf{Cont}(\mathbf{cod}) = \mathbf{Fam}(\mathbf{cod}^{\text{op}})$ lives on the *opposite*, lens side of the same theory (the reverse-derivative side, [4, Ex. 32, Def. 57]). The naive hope “the container bifibration *is* the forward tangent fibration” is therefore false, and false for a precise reason: the difference between pullback reindexing ρ^* and $(\Sigma_\rho)^{\text{op}}$. The role of [4] here is a packaging remark that re-delivers the differential category we have already built; nothing in the proof rests on it.

Does it see categories? A directed container is the same thing as a small category [2]; in polynomial terms, a small category is a $\triangleleft$ -comonoid. So the natural follow-up is whether T respects this comonoid structure. In §6 we record that it does, in a precise and slightly surprising sense: T is a *strict $\triangleleft$ -monoidal base-change functor* $\mathbf{DCont}(k) \rightarrow \mathbf{DCont}(\mathbb{D})$, where $\mathbb{D} = k[\varepsilon]/\varepsilon^2$; the monoidal comparison is the container chain rule, holding on the nose. Hence T sends a small category to a small category — but over the dual numbers, not over the original base. This is not a defect of the construction; we show the obstruction to keeping it over $\mathbf{Set}$ is exactly the second-order term that $\varepsilon^2 = 0$ sends to the zero morphism. The headline is clean: *the tangent bundle of a category lives over the dual numbers, a tangent vector at an object a is a morphism out of a , and the zero vector is the identity*.

Outline. §2 fixes containers, the category $\mathbf{SPoly}$, the differential-category axioms we use, and the counting functor. §3 proves the main theorem by reflection and unpacks the combinatorial content of the three non-trivial axioms. §4 proves the map-level linearity discriminator. §5 positions the result in the fibrational theory and states the orientation hazard. §6 records the restriction to directed containers and its honest caveat. §7 states novelty honestly and lists the open edges.

2 Preliminaries

We recall only what the proof uses.

Containers and polynomial functors. A *container in n variables* is a set S of *shapes* together with, for each shape $s \in S$ and each $i \in \{1, \dots, n\}$, a *position set* P_s^i . Its extension is the polynomial functor

$$\llbracket S \triangleleft P \rrbracket : \mathbf{Set}^n \rightarrow \mathbf{Set}, \quad \vec{X} \mapsto \sum_{s \in S} \prod_{i=1}^n X_i^{P_s^i}.$$

We work with *finite-support* containers: finitely many shapes ($|S| < \infty$), each with finitely many positions (all P_s^i finite). Thus only the finite cardinalities $|P_s^i|$ and the finite shape set matter up to isomorphism; this is what makes the counting functor N (Definition 3) land in $\mathbb{N}$ — for an infinite-shape container such as $\mathbb{N} \cdot y$ the count would be $\aleph_0 \cdot m \notin \mathbb{N}$. Composition of containers is substitution of one extension into another, written $G \triangleleft F$; it corresponds to functor composition [1].

The base category.

Definition 2 ($\mathbf{SPoly}$). Let $\mathbf{SPoly}$ be the category with objects the natural numbers $n \in \mathbb{N}$ (read n as “ n variables”, i.e. $\mathbf{Set}^n$); morphisms $n \rightarrow m$ the isomorphism classes of m -tuples of finite-support containers in n variables (equivalently, the finite-support polynomial functors $\mathbf{Set}^n \rightarrow \mathbf{Set}^m$); composition substitution; identities the variable tuples $(y_1, \dots, y_n)$.

Passing to isomorphism classes makes substitution strictly associative and unital, so $\mathbf{SPoly}$ is a genuine 1-category. Finite products are $n \times n' = n + n'$ with the variable-projection tuples as projections and 0 terminal. Each hom-set $\mathbf{SPoly}(n, m)$ is a commutative monoid under coproduct of containers $+$ (unit the empty container 0), and composition is left-additive, $(f + g) \circ h = f \circ h + g \circ h$ and $0 \circ h = 0$, because an outer coproduct distributes over any substitution on the right. With products and $(+, 0)$ computed componentwise and compatibly, $\mathbf{SPoly}$ is a *Cartesian left-additive category* in the sense of [3, Def. 1.2.1].

The differential combinator. For $f: n \rightarrow m$ define $\partial[f]: 2n \rightarrow m$ by the directional one-hole derivative

$$\partial[f](\vec{x}; \vec{v}) := \sum_{i=1}^n v_i \cdot \partial_i f,$$

where $\partial_i f$ is the AAGM one-hole derivative in the i -th variable: a shape s with $k = |P_s^i|$ positions in variable i contributes k shapes, each replacing one of those positions by a *new* tangent variable v_i and leaving the other $k - 1$ as x_i . Thus $\partial[f]$ is the $2n$ -variable container that is degree 1 in the tangent variables. The AAGM endofunctor ∂ is the v -linear part of $\partial[-]$, i.e. $\partial[-]$ on $\text{SPoly}(1, 1)$.

The differential-category axioms we need. A *Cartesian differential category* [3] is a Cartesian left-additive category with a combinator D taking $f: A \rightarrow B$ to $D[f]: A \times A \rightarrow B$ (thought of as f 's directional derivative, linear in the second argument), subject to seven axioms CD.1–CD.7. We will invoke, by name, only the three with combinatorial content:

- **CD.5 (chain rule)** $D[g \circ f] = D[g] \circ \langle f \circ \pi_1, D[f] \rangle$;
- **CD.6** $D[D[f]] \circ \langle \langle a, b \rangle, \langle 0, d \rangle \rangle = D[f] \circ \langle a, d \rangle$;
- **CD.7 (symmetry of mixed partials)** $D[D[f]] \circ \langle \langle a, b \rangle, \langle c, 0 \rangle \rangle = D[D[f]] \circ \langle \langle a, c \rangle, \langle b, 0 \rangle \rangle$.

A map f is *linear* when $D[f] = f \circ \pi_2$ (its derivative is itself, independent of the base point); the linear maps are the morphisms along which the differential is additive. Every Cartesian differential category is a Cartesian tangent category, with tangent functor $T(A) = A \times A$, $T(f) = \langle f \circ \pi_1, D[f] \rangle$, projection $p = \pi_1$ and zero section $\langle 1, 0 \rangle$ [5, Prop. 4.7].

The counting functor.

Definition 3 ($\text{Poly}_{\mathbb{N}}$ and the counting functor N). Let $\text{Poly}_{\mathbb{N}}$ be the Cartesian differential category of polynomials over the rig $\mathbb{N}$: objects $n \in \mathbb{N}$, morphisms $n \rightarrow m$ the m -tuples of polynomial functions $\mathbb{N}^n \rightarrow \mathbb{N}$ with $\mathbb{N}$ -coefficients, composition substitution, differential the Jacobian directional derivative $D[p](\vec{x}; \vec{v}) = \sum_i v_i \partial p / \partial x_i$. (Polynomials over any commutative rig form a Cartesian differential category — the motivating example of [3] — the axioms using only $+$, $\times$, d/dx and never subtraction.) Define $N: \text{SPoly} \rightarrow \text{Poly}_{\mathbb{N}}$ on objects by $N(n) = n$ and on a container $F = S \triangleleft P$ by its *counting polynomial*

$$N(F)(m_1, \dots, m_n) = \sum_{s \in S} \prod_{i=1}^n m_i^{|P_s^i|} \in \mathbb{N}[m_1, \dots, m_n].$$

Dual numbers and directed containers. Write $\mathbb{D} = k[\varepsilon]/\varepsilon^2$ for the dual numbers over a base rig/ring k ; these appear only in §6. A *directed container* is, by the theorem of [2], exactly a small category C , presented as the polynomial comonad

$$c(y) = \sum_{a \in \text{Ob } C} y^{\text{Out}(a)}, \quad \text{Out}(a) = \{\text{morphisms with domain } a\},$$

with counit picking id_a and comultiplication encoding codomain and composition; the comonoids in $(\text{Poly}, \triangleleft, y)$ are precisely the small categories.

3 The theorem and its proof

Theorem 4. *SPoly is a Cartesian differential category with differential combinator $D = \partial$. Hence (SPoly, ∂) is a Cartesian tangent category with $T(n) = 2n$, $T(f) = \langle f \circ \pi_1, \partial[f] \rangle$, realised by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$.*

The proof is three lemmas: N preserves the structure, N is faithful, faithful functors reflect the axioms.

Lemma 5 (N preserves the differential-category data). *N is a product-preserving, left-additive functor commuting with the differential combinators: $N(\partial_i F) = \partial N(F)/\partial m_i$, and hence $N(\partial[f]) = D[N(f)]$.*

Proof. Each clause is a counting identity. *Composition:* for set-valued polynomial functors the cardinality of a composite is the composite of cardinalities, $|\llbracket G \triangleleft F \rrbracket(\vec{X})| = |\llbracket G \rrbracket(\llbracket F \rrbracket(\vec{X}))|$, and $|\llbracket G \rrbracket(-)|$ depends only on the cardinality of its argument; so $N(g \circ f) = N(g) \circ N(f)$. *Products, projections, identities:* $\llbracket - \rrbracket$ sends $n \times n' = n + n'$ to the product of counting rings and variable-projections to coordinate projections. *Left-additivity:* coproduct of containers $\mapsto$ sum of counting polynomials, $0 \mapsto 0$. *Products of functors:* $|F \times G| = |F| \cdot |G|$, so $\times \mapsto \times$. *Derivative:* $\partial_i F$ replaces one of the $|P_s^i|$ positions in variable i by a tangent copy, with counting

$$\sum_s |P_s^i| m_i^{|P_s^i|-1} \prod_{j \neq i} m_j^{|P_s^j|} = \frac{\partial}{\partial m_i} N(F);$$

summing $v_i \cdot (-)$ over i gives $N(\partial[f]) = D[N(f)]$. □

Lemma 6 (N is faithful on isomorphism classes). *For parallel $f, g: n \rightarrow m$ in SPoly, $N(f) = N(g)$ implies $f = g$.*

Proof. It suffices to treat $m = 1$. The monomials of $N(F) = \sum_s \prod_i m_i^{|P_s^i|}$ are $\prod_i m_i^{k_i}$, and the coefficient of $\prod_i m_i^{k_i}$ counts the shapes s with exponent vector $(|P_s^1|, \dots, |P_s^n|) = (k_1, \dots, k_n)$. So the formal polynomial $N(F)$ determines the multiset of exponent vectors, which is the isomorphism class of F (shapes up to bijection, positions up to cardinality). Over the infinite rig $\mathbb{N}$ a polynomial function determines its formal polynomial, so N is injective on isomorphism classes. □

Lemma 7 (Reflection). *Each differential-category axiom holds in SPoly.*

Proof. Fix an axiom: an equation $L = R$ of parallel morphisms built from a map f together with projections, tupling, $+$, 0 , $\times$, $\circ$ and $\partial[-]$. By Lemma 5 N preserves every one of these operations, so $N(L)$ and $N(R)$ are the two sides of the *same* axiom instantiated in $\text{Poly}_{\mathbb{N}}$ at $N(f)$. Since $\text{Poly}_{\mathbb{N}}$ is a Cartesian differential category (Definition 3), $N(L) = N(R)$. By Lemma 6 N is faithful, and a faithful functor reflects equalities of parallel morphisms; hence $L = R$ in SPoly. □

This proves the differential-category claim of Theorem 4. The tangent structure is the canonical one attached to any Cartesian differential category [5, Prop. 4.7]; under N it reads $N(f)(x + \varepsilon v) = N(f)(x) + \varepsilon \sum_i v_i \partial_i N(f) \bmod \varepsilon^2$, i.e. the dual-number substitution, which Lemmas 5–7 show computes T on the nose in SPoly.

Example 8 (The worked monomial). For $p = y^n$ the dual-number substitution gives $(y + \varepsilon v)^n = y^n + \varepsilon n v y^{n-1} \bmod \varepsilon^2$, so the ε -coefficient is $n \cdot y^{n-1} = \partial(y^n)$: the one-hole contexts of an n -tuple, recovered as a tangent vector. The base part y^n is the projection $p \circ \pi_1$. For $n = 1, 2, 3$ this is, respectively, v ; $2vy$; $3vy^2$ — the n ways to be a one-hole context.

What reflection hides. Lemma 7 certifies every axiom but suppresses the bijections. The three with content are worth stating as container isomorphisms, both because they are the mathematics and because they are the parts a referee will want to see by hand.

CD.5, the chain rule. At container level this is $\partial(G \triangleleft F) \cong (\partial G \triangleleft F) \times \partial F$, proved directly as an explicit bijection in [7, Thm. 1]: a hole in the composite is a choice of outer G -index together with a hole in the chosen inner block; the leftover splits as “some other inner block” ($\partial G \triangleleft F$) and “this inner block” (∂F). Lemma 7 re-derives it as a corollary of the classical chain rule plus faithfulness — a consistency check on that bijection.

CD.6. Write $f = S \triangleleft P$ in one variable. $D[D[f]]$ has two shape families: *Case B*, where the second derivative pokes a base position (two distinct holes, carrying both tangent directions), and *Case T*, where it pokes the existing tangent position (one hole). Setting the second base slot to 0 annihilates every Case-B shape, leaving Case T, whose container is exactly $D[f]$ with the surviving tangent direction; this is the required identity.

CD.7, Clairaut for containers. Setting the second tangent slot to 0 leaves only Case B: shapes (s, h_1, h_2) with distinct holes $h_1 \neq h_2 \in P_s$ carrying tangent directions b, c . The swap involution $(s, h_1, h_2) \mapsto (s, h_2, h_1)$ is a bijection of ordered distinct hole-pairs exchanging the two directions, witnessing symmetry of the mixed second derivative. In several variables a hole is a pair (variable i , position in P_s^i) and the same swap applies.

Corroboration. The single-variable axioms CD.1–CD.7 were checked symbolically with an *abstract* f (hence for all single-variable containers, N being faithful), and $N(\partial_i F) = \partial_i N(F)$, the chain rule, and the CD.7 swap on several thousand random multivariable finite containers. A minimal Mathlib-free Lean 4 development certifies the core computable facts — the ε -coefficient identity of Example 8 for $n \leq 5$, the monomial Leibniz and chain-rule shadows, and the map-level linearity discriminator of §4 — with no axioms and no appeal to `native_decide`. These are finite oracles corroborating the reflection, not the reflection itself.

4 Linearity: the corrected discriminator

There is a reading of “differential structure on containers” at the level of *objects* that is tempting and vacuous. One can ask which objects B are linear in the sense $TB \cong B \times B$; but $\mathbf{SPoly}$ is a *full* Cartesian differential category, so every object is linear and the question is empty (in the language of [4, Thm. 69], $\text{Diff}(T)$ is everything). The non-vacuous statement is one level down.

Proposition 9 (Linear maps). *A morphism $f : 1 \rightarrow 1$ is a linear map of the Cartesian differential category — equivalently an additive container, equivalently $\partial[f](x, v) = f(v)$ — if and only if $f = S \cdot y$ for a set S (a linear container). In particular $y^2, y^3, \dots$ and nonzero constants are not linear.*

Proof. f is linear iff $D[f] = f \circ \pi_2$. Apply N : $v \cdot f'(x) = f(v)$ for all x, v as $\mathbb{N}$ -polynomials. Writing $N(f) = \sum_k a_k x^k$, the left side is $\sum_k a_k k v x^{k-1}$ and the right $\sum_k a_k v^k$; equality forces $a_k = 0$ for $k \neq 1$, i.e. $N(f) = a_1 x$. By Lemma 6, $f \cong a_1 \cdot y = S \cdot y$ with $|S| = a_1$. Conversely $D[S \cdot y] = S \cdot v = (S \cdot y)(v)$. $\square$

This is the honest non-vacuity certificate. Under the trivial candidate $T(p) = p \times p$ every map is linear; Proposition 9 shows the actual tangent functor is not that — y is linear and y^2 is not. Table 1 is the summary to carry away.

container f	$N(f)$	$\partial[f](x, v)$	linear?
y	x	v	yes
$S \cdot y$	$ S x$	$ S v$	yes
y^2	x^2	$2vx$	no
y^3	x^3	$3vx^2$	no
constant c	$ c $	0	no (unless $c = 0$)
affine $c + S \cdot y$	$ c + S x$	$ S v$	no (unless $c = 0$)

Table 1: The map-level linearity discriminator. Linear $\iff \partial[f](x, v) = f(v) \iff f = S \cdot y$. Machine-checked, zero axioms, in the Lean development.

5 The orientation split: a fibrational positioning

The fibrational theory of first-order differential structures [4] gives a common home to differential categories, tangent categories and their reverse variants. Its master notion: a first-order differential structure on an object B (of a suitable 2-category) is a fibration with fibred biproducts together with a *section* of it; the tangent-category instance [4, Thm. 56] takes the fibration to be the *additive codomain fibration* — the codomain fibration $\text{cod}: B^\rightarrow \rightarrow B$, whose Cartesian maps are pullbacks, refined so each fibre object is a commutative monoid in the slice — and a section T sends each object to a bundle over it. Their Theorem 74 then shows a first-order Cartesian tangent structure of this kind induces a first-order Cartesian differential category, with combinator $D_T = t^{-1} \circ T f \circ t$.

Our dual-number T realises exactly this forward picture: it is a section of the additive codomain fibration, and the induced combinator D_T is ∂ . In this sense Theorem 74 *re-delivers* the differential category of §3. We stress that this is a packaging remark: nothing in the proof of Theorem 4 depends on [4], and the citation is here to locate the result in its natural taxonomy, not to carry it.

The hazard. There is a seductive stronger claim — that the container bifibration itself *is* the forward tangent fibration of [4] — and it is false. The container logic $\text{Cont}(\text{cod}) = \text{Fam}(\text{cod}^{\text{op}})$ is built on cod^{op} , and [4, Ex. 32] identifies cod^{op} of Set with dependent lenses, i.e. polynomial functors, i.e. containers; the *reverse* differential side of that theory uses the dual of the additive-bundle fibration [4, Def. 57]. So $\text{Fam}(\text{cod}^{\text{op}})$ sits on the reverse/lens side, not the forward Thm-56/74 side. The distinction is the difference between pullback reindexing ρ^* and the opposite direct-image $(\Sigma_\rho)^{\text{op}}$ — a sign error at the level of fibrations. We name it explicitly because it is the same orientation hazard that has mis-oriented adjacent results in this programme more than once; the safe statement is the forward one, on cod with ρ^* , which is what Theorem 4 uses.

Remark 10. The reverse/lens reading — “is ∂ a reverse-derivative tangent structure on $\text{Fam}(\text{cod}^{\text{op}})$?” — is a genuine and, as far as we know, open question. We do not address it here; see §7.

6 Does the tangent structure see categories?

A directed container is a small category, i.e. a $\triangleleft$ -comonoid [2]. If T respects that comonoid structure it transports small categories to small categories. We record the expected statement as a *sketch*: the monoidal comparison is verified, but the final comonoid step is only computed, not proved in full, so we state it as a proposition-sketch rather than a theorem. The twist in it is the whole point.

Proposition 11 (Restriction to directed containers — sketch). *T is a strict $\triangleleft$ -monoidal functor*

$$T: (\text{Poly}/k, \triangleleft, y) \longrightarrow (\text{Poly}/\mathbb{D}, \triangleleft_{\mathbb{D}}, y + \varepsilon v), \quad \mathbb{D} = k[\varepsilon]/\varepsilon^2,$$

where $\triangleleft_{\mathbb{D}}$ is substitution after base change along $k \rightarrow \mathbb{D}$; the monoidal comparison is the container chain rule $\partial(G \triangleleft F) \cong (\partial G \triangleleft F) \times \partial F$, holding on the nose (laxator the identity). Granting the standard base-change 2-functoriality of polynomial composition [6], T should then preserve $\triangleleft$ -comonoids, sending a small category internal to k to one internal to $\mathbb{D}$: a base-change functor $\text{DCont}(k) \rightarrow \text{DCont}(\mathbb{D})$. We have verified the strict $\triangleleft$ -monoidality; the comonoid-preservation step is computed on examples but not proved in general, which is why this is a sketch.

The verified content is a single identity. Writing $T(c)(y, v) = c(y) + \varepsilon v c'(y)$, define $\triangleleft_{\mathbb{D}}$ by substituting the dual-number output of $T(d)$ into c ; then $T(c \triangleleft d)$ has tangent part $v(c \circ d)'(y)$, and the ordinary chain rule $(c \circ d)' = c'(d) \cdot d'$ is exactly the $\triangleleft_{\mathbb{D}}$ tangent part. So T is strict $\triangleleft$ -monoidal, and the AAGM container chain rule (§3) is this identity read off in the (shape $\triangleleft$ position) presentation. Comonoid preservation *would* follow because strict monoidal functors preserve comonoids — but this is the step we do not re-derive on container morphisms, and we flag it as the conjectural part, borrowing in the examples the base-change 2-functoriality of [6].

Remark 12 (Why the base is $\mathbb{D}$ and not k — the honest caveat). T is *not* an endofunctor of $\text{DCont}(k)$. Over Set the polynomial $T(c) = c + \varepsilon v \partial c$ carries no comonad structure restricting to C : composition is not even total. Present $T(c)$ as a would-be two-sorted directed container, with base objects $B(a)$ (a faithful copy of C) and tangent objects $Tn(a, f)$ indexed by the morphisms f of C , the derivative ∂c giving each $Tn(a, f)$ the out-positions $\{\star\} \cup (\text{Out}(a) \setminus \{f\})$. The counit forces $\text{id}_{Tn(a, f)} = \star$, and the sort constraints force cod and composition — except that the composite $\text{comp}_C(g, r)$ of a surviving position $g \in \text{Out}(a) \setminus \{f\}$ with a base position r can equal the *removed* position f , in which case the required composite has no target among $Tn(a, f)$'s positions. For any category with a non-identity morphism $f \neq \text{id}_a$ the pair $g = \text{id}_a, r = f$ exhibits this, so the Set reading fails universally outside the discrete/terminal cases (verified by complete enumeration on the walking arrow, $\mathbb{Z}/2$, and an idempotent monoid). The resolution is the dual numbers: that missing composite re-enters the differentiated direction, producing an ε^2 term, which $\varepsilon^2 = 0$ sends to the *zero morphism* — not a position any Set -container can carry, but a legitimate morphism of a $\mathbb{D}$ -module-enriched category. This is exactly why $\triangleleft_{\mathbb{D}}$ (base change), and not free two-variable substitution, is the correct composition.

The geometric payoff survives the caveat and is clean. The projection $p: T(C) \rightarrow C$ (base change $\mathbb{D} \rightarrow k, \varepsilon \mapsto 0$) has fibre over a the out-morphisms $\text{Out}_C(a)$, and the zero section $0: C \rightarrow T(C)$ lands on the identity direction. So:

A tangent vector of a category at an object a is a morphism out of a ; the zero tangent vector is the identity id_a ; and the tangent bundle lives over the dual numbers.

Problem 13 (open). Give the closed-form multiplication of $T(C)$ as a $\mathbb{D}$ -linear category, via $\delta_{T(c)} = \mu^{-1} \circ T(\delta_c)$ with T defined on container *morphisms* over $\mathbb{D}$ (so the ε -zero morphism is representable), and identify it with a named construction — a tangent algebroid / first-order jet category, or C glued to its arrow-bimodule by the ε -action. The object counts already exclude the obvious Set candidates (for the walking arrow, the arrow category has 3 objects, $C \otimes \mathbb{D}$ has 4, and $T(C)$ has 5).

7 Honest novelty, and the open edges

What is and is not new. The derivative ∂ is [1]; the chain rule under $\triangleleft$ is [7]. That set-valued polynomial functors assemble into a Cartesian differential category with ∂ as combinator is, to our knowledge, not recorded in exactly this form, but it is close to folklore: it is the **Set**-coefficient instance of “polynomials over a rig form a Cartesian differential category” [3], and it is adjacent to the differential structure of Joyal’s species and analytic functors. We do not claim the differential category as deep. The contribution is the clean statement, the reflection proof (which makes the seven axioms inherited rather than re-checked), the resolution of the object-versus-map confusion (Proposition 9), the fibrational positioning with its named orientation hazard (§5), and the restriction to directed containers with the dual-number caveat made precise (§6).

Open edges, stated honestly. (i) *First order only.* Both ∂ and the forward theory of [4] are first order; $\varepsilon^2 = 0$ truncates the Faà di Bruno tower. Higher and iterated derivatives are out of scope. (ii) *The reverse/lens reading is unaddressed.* Whether ∂ is a reverse-derivative tangent structure on $\mathbf{Fam}(\mathbf{cod}^{\text{op}})$ — the side on which the container logic actually lives — is open; §5 only explains why it is a different question. (iii) *The closed form of $T(C)$* is Problem 13: we have the underlying object, the projection, the zero section and the geometry, but not the named multiplication. (iv) *An endofunctor tangent structure “on $\mathbf{DCont}$ ”* would come, if at all, from the Weil-algebra-indexed family $W \mapsto \mathbf{DCont}(W)$ with base-change functors (Leung’s presentation of tangent structure as an action of Weil algebras), of which $\mathbb{D}$ is the first infinitesimal; this is sketched, not proved.

Why it matters. For the differential- and tangent-category reader, the result supplies a concrete, combinatorial model whose differential combinator is a construction from programming-language theory, with a one-line discriminator (Table 1) separating it from the trivial tangent functor. For the container and **Poly** reader, it upgrades the derivative from mnemonic to structure and shows precisely how far that structure reaches into the directed (categorical) world — as far as the dual numbers, and no further over **Set**. The headline is portable: *the tangent bundle of a category lives over the dual numbers, the one-hole derivative is its vertical part, and the obstruction to keeping it over **Set** is the $\varepsilon^2 = 0$ zero morphism.*

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